SASTRA DEEMED TO BE UNIVERSITY
School of Computing — Department of Computer Science & Engineering
TIRUMALAISAMUDRAM, THANJAVUR – 613401, TAMIL NADU, INDIA

ACADEMIC INTEGRITY AND AI-TOOL USAGE DECLARATION

Project Title: AUTOSAR AI – High-Level Design Document Intelligence Platform

Student Name:	Hamdhan Ahmed B
Registration Number:	127003081
Degree & Branch:	B.Tech. Computer Science & Engineering (AIML)
Repository URL:	https://github.com/hamdhan05/AUTOSAR-AI-Implementation

SECTION A — ACADEMIC INTEGRITY DECLARATION

I, Hamdhan Ahmed B (Registration Number: 127003081), a bonafide student of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence and Machine Learning) at SASTRA Deemed to be University, hereby solemnly declare that the capstone project entitled **"AUTOSAR AI – High-Level Design Document Intelligence Platform"** is an original, independent record of technical and research work undertaken by me under academic supervision.

I formally certify and affirm adherence to the following core academic integrity principles:
1. Originality of Submission: The architectural design, software services, database models, prompt engineering structures, and analytical workflows presented in this project are the original work of the student, except where specifically cited and acknowledged.
2. Appropriate Acknowledgment of External Assets: All third-party libraries, frameworks (FastAPI, Streamlit, PyMuPDF, SQLAlchemy, ChromaDB), pre-trained models (SentenceTransformers all-MiniLM-L6-v2, Ollama llama3.2:3b), and documentation standards (AUTOSAR, ISO 26262) have been formally recognized, attributed, and referenced.
3. Zero Fabrication of Test Results: All reported empirical outcomes—including the 28 automated unit and integration tests passing with a 100% pass rate in 27.35 seconds on October 13, 2026, and the 35-query benchmark evaluation—are genuine, reproducible, and verifiable directly from the repository source code without artificial score invention.
4. Confidentiality & Synthetic Data Compliance: No proprietary, classified, or trade-secret

automotive OEM source code or specifications have been utilized or disclosed. All sample HLD documents are synthetic and developed strictly for academic demonstration.

5. Student Verification & Sole Responsibility: The final submission, codebase, documentation, and technical report have been reviewed by me, and I assume full academic and intellectual responsibility for their correctness, validity, and rigor.

Place: Thanjavur, Tamil Nadu
Date: October 13, 2026

Signature of the Student:

_____

Hamdhan Ahmed B (Reg No: 127003081)

SECTION B — AI-TOOL USAGE DECLARATION

In accordance with institutional academic integrity policies and emerging international guidelines on the ethical adoption of artificial intelligence in higher education, the table below documents all AI models, embedding representations, development assistants, and automated evaluation tools utilized throughout the lifecycle of this project. Each tool is clearly differentiated by its operational role—runtime inference, vector embedding, development assistance, or automated verification.

AI Model/Tool	Purpose	Artifact Affected	Verified by Student
Ollama with llama3.2:3b (Meta Llama 3.2 3B Quantized)	Local generative LLM inference for grounded engineering question-answering in the AI Assistant runtime service. Operates 100% on-premises via local HTTP REST (localhost:11434).	backend/services/llm_service.py frontend/pages/4_AI_Assistant.py Runtime QA responses and citation formatting	Verified by Student (Live HTTP probe and abstention test passing; grounded prompt active)
SentenceTransformers (all-MiniLM-L6-v2)	Generating 384-dimensional dense semantic vector embeddings for document chunks during ingestion and nearest-neighbor cosine similarity retrieval.	backend/services/vector_store_service.py ChromaDB persistent collections (project_{id}) Semantic search queries	Verified by Student (Automated ingestion tests passing; persistent vector storage validated)
Google Gemini / Antigravity AI Coding Assistant	Assisted development workflow: code scaffolding, test case template drafting, technical documentation compilation, presentation scripting, and formatting assistance.	Backend router boilerplate, automated test scripts in tests/unit/, documentation drafts, presentation scripts	Pending student verification (Subject to student line-by-line review, compilation, and viva defense)
Pytest 9.1.1 Automated Test Runner	Automated test execution, assertion checking, and regression verification across 28 unit and integration test cases spanning 12 subsystem modules.	tests/unit/ test suite Evaluation_Results/Test_Results.md Automated pass rate verification	Verified by Student (All 28 tests independently executed and verified passing on Oct 13, 2026)

Important Operational Distinctions & Academic Governance Notes:
1. Runtime vs. Development AI: A fundamental distinction is maintained between runtime neural models (Ollama llama3.2:3b and SentenceTransformers), which execute locally as core components of the deployed software architecture, and development assistants (Google Gemini / Antigravity), which were utilized as software engineering accelerators during implementation.
2. Student Ownership of AI-Assisted Artifacts: All code snippets, configurations, test fixtures, and documentation drafts suggested by AI coding assistants have been scrutinized, debugged, executed, and validated by the student. Any item marked as 'Pending student verification' indicates that the student personally re-audits the artifact before final capstone evaluation.
3. Sole Academic Accountability: In alignment with university guidelines, AI tools are acknowledged strictly as assistive technological instruments. The academic, scientific, and technical integrity of the entire project remains the sole responsibility of the submitting student.

FACULTY ENDORSEMENT & UNIVERSITY EXAMINER APPROVAL

Project Guide Signature:

_____

Name:
Designation:
Department of Computer Science & Engineering

Head of the Department Signature:

_____

Head of Department
School of Computing, SASTRA University

Internal Examiner Signature:

_____

Name:
Date:

External Examiner Signature:

_____

Name:
Date:

Candidate Signature:

_____

Hamdhan Ahmed B (Reg No: 127003081)

Academic Institution Seal:

[School of Computing]
SASTRA Deemed University